Prob 1

$$u_t = u_{xx}$$

$$u(x+h,t) = u(x,t) + u'(x,t)h + \frac{u''(x,t)h^2}{2}$$

$$u(x-h,t) = u(x,t) - u'(x,t)h + \frac{u''(x,t)h^2}{2}$$

$$u(x+h,t) + u(x-h,t) = u(x,t) + u'(x,t)h + \frac{u''(x,t)h^2}{2} - \left(u(x,t) - u'(x,t)h + \frac{u''(x,t)h^2}{2}\right)$$

$$u''(x,t) = \frac{u(x-h,t) - 2u(x,t) + u(x+h,t)}{h^2} = \frac{U_{m-1}^n - 2U_m^n + U_{m+1}^n}{h^2}$$

$$u(x,t+k) = u(x,t) + u'(x,t)k$$

$$u(x,t-k) = u(x,t) - u'(x,t)k$$

$$u(x,t+k) - u(x,t-k) = u(x,t) + u'(x,t)k - (u(x,t) - u'(x,t)k)$$

$$u'(x,t) = \frac{u(x,t+k) - u(x,t-k)}{2k} = \frac{U_m^{n+1} - U_m^{n-1}}{2k}$$

$$u_t = u_{xx} \Rightarrow \frac{U_m^{n+1} - U_m^{n-1}}{2k} = \frac{U_{m-1}^n - 2U_m^n + U_{m+1}^n}{h^2} \Rightarrow U_m^{n+1} = \frac{2k}{h^2}\left(U_{m-1}^n - 2U_m^n + U_{m+1}^n\right) + U_m^{n-1}$$

Prob 2

$u_t = u_{xx} \quad , \quad u(x,0) = e^{-2x^2} \quad \Rightarrow \quad \hat{u}(\omega,0) = \frac{1}{2} e^{-\frac{\omega^2}{8}}$

$\mathcal{F}(u_t) = \mathcal{F}(u_{xx})$

$\hat{u}_t = -\omega^2 \hat{u}$

$\hat{u}_t + \omega^2 \hat{u} = 0$

$d\hat{u}$